

LUNDS TEKNISKA HÖGSKOLA
Inst. för telekommunikationssystem

Tentamen i *DATORKOMMUNIKATION*

Datum: 2005-12-15

Tid: 14 – 19

For tasks 1-5: you must NOT use the text book. For tasks 6-8: you may use the text book

SOLUTIONS!

The total exam is judge as 3: 50-64 cr, 4: 65-79 cr, 5: 80 cr or more.

TASKS 1-5, the text book must not be used!

Task 1 (10 cr)

- a) The physical layer is responsible for transmitting a bit stream over a physical medium. It is concerned with
 - physical characteristics of the media
 - representation of bits
 - type of encoding
 - synchronization of bits
 - transmission rate and mode
 - the way devices are connected with each other and to the links
- b) Parity check, cyclic redundancy check, and checksum.
- c) A root bridge is the bridge with the smallest ID. A designated bridge has the least cost path between the LAN and the root bridge.

Task 2 (10 cr)

- a) To provide flow and error control.
- b) In Selective Repeat ARQ, just the damaged frame is re-transmitted. In Go-Back-N ARQ, all frames starting with the damaged frame are re-transmitted.
- c) The sender sets a timer when a frame is sent. If an acknowledgment is not received by the time the timer expires, the frame is re-transmitted.
- d) Go-Back-N ARQ and Selective Repeat ARQ.
- e) The bandwidth delay product is a measure of the number of bits that can be sent out while waiting for news from the receiver.

Task 3 (10 cr)

- a) LLC sublayer: responsible for creating a reliable connection over a link. The LLC sublayer uses error detection mechanisms and acknowledgements.
MAC sublayer: responsible for the principles for accessing the media. The MAC sublayer examples: CSMA/CD, Token passing, Polling.
- b) CSMA/CD is for multi access networks like Ethernets. A station the wants to send first senses whether the channel is busy or not (Carrier Sense). If busy the station defers. The station then takes actions dependent of whether it operates under a persistent or non-persistent mechanism. If the medium is free the station sends. After having sent a packet, the sending station waits for an acknowledgement from the receiver. If an acknowledgement is not received before the senders timer has experienced, a retransmission of the packet is done. During sending, the station

listens to the medium and if it detects a collision, it stops sending the packet, but sends out a short jam signal.

- c) Under Controlled based MAC protocols there is a mechanism that distributes the rights to send to one station at a time. This mechanism can be either centralized (polling) or decentralized (token passing). Thus no collisions can occur. Under Random Access based MAC protocols the different stations compete for the media. Collisions may occur.
- Controlled based MAC protocols work better at high loads. Random Access based MAC protocols work better at low loads.
 - Controlled based MAC protocols may give QoS guarantees, whereas for Random Access based MAC protocols this is much harder to achieve.

Task 4 (10 cr)

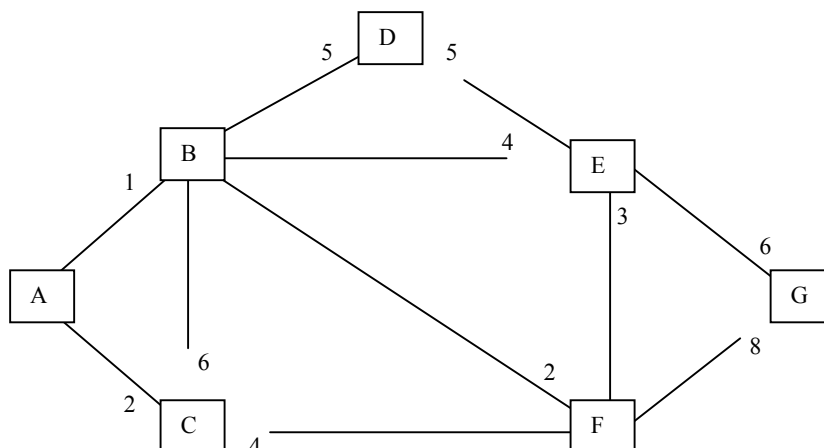
- a) The Internet Protocol (IP) is unreliable in the sense that it does not provide any guarantees at all. It is a connectionless datagram protocol, which means that IP does not set up an end-to-end connection, each IP frame is handled separate from all other IP frames also within one and the same session. It is a best-effort delivery service in the sense that it provides no error control or flow control.
- b) Larger address space, Better header format, New options, Allowance for extensions, Support for resource allocation, Support for more security. (You just have to mention three of these).
- c) TCP is reliable in the sense that it provides error control. The three most important functions of the TCP protocol are error control, flow control and the guarantee of no duplicate frames nor frames out of order.

Task 5 (10 cr)

a) Dijkstra Algorithm

1. Select the root and make it permanent.
2. Add neighbour arcs and nodes. Note cumulative cost to these nodes. Make them tentative. If there are more than one arc to a node select the arc with least cumulative cost.
3. Make tentative node with least cumulative cost permanent.
4. Repeat 2 and 3 until all nodes are permanent.

b)



c)

B	1	-
C	2	-
D	5	B
E	3	B
F	2	B
G	6	B

TASKS 6-8, the textbook may be used!

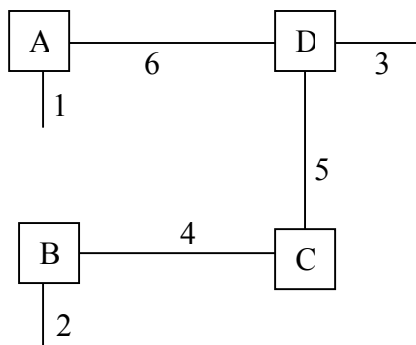
Task 6 (15 cr)

The first frame carries DNS (destination port 53) over UDP over IP over Ethernet II. It is destined for node C, so this has to be the DNS server.

The second frame has a destination IP address that is not belonging to the same subnet as A (182.3.6.72 is a class B address). From the MAC destination address it is concluded that node D must be the Def-GW.

Task 7 (15 cr)

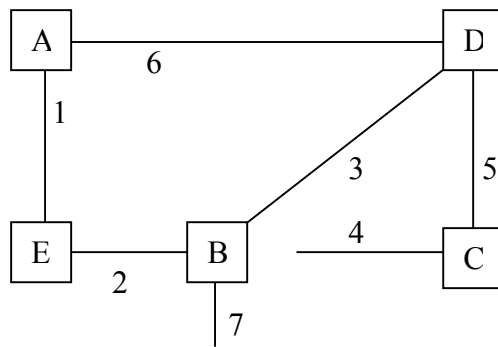
a)



b)

Router A routing table		
Subnet	Cost	Next-hop
1	1	-
2	2	E
3	2	D
4	3	D
5	2	D
6	1	-
7	3	E

c)



Task 8 (10 cr)

- a) Repeater? No! A repeater just reshapes the signals on one of its sides and sends it out on the other side. Bridge? Yes! A bridge must have both layer 1 and 2 functionalities for the two sides.
- b) The part of the router facing the 10Base-T Ethernet. Layer 1: transceiver for 10Base-T connecting to the twisted pair, Layer 2: CSMA/CD and LLC, Layer 3 IP. The part of the router facing the Token passing network. Layer 1: transceiver for the token passing network, that provides all the necessary means. Layer 2: The token passing MAC protocol and LLC, Layer 3 IP.
- c) The physical and data link layers are of concern to LANs.